Blocker:
Non-linear representations in quantum mechanics

This file collects everything about the one item of `open-problems.tex` that blocks verification, together with the author's replies, a proposed resolution and the changes it needs in the brief. Sections 1 and 2 are copied from `open-problems.tex` and `formalization-log.tex`; the rest is new.

1 The blocker

1. [BLOCKS VERIFICATION: BRIEF ISSUE] **Theorem 1 as an independence statement.** "REP-VSLIN is independent of QST" is a statement about logical independence, but ¬REP-VSLIN is not made precise, and its truth depends on the reading.

   - *Literal independence over models.* If a system is only its state space and probability, REP-VSLIN follows from QST (take the H that QST provides), so ¬REP-VSLIN is inconsistent with QST. If a system also carries vector-level data that may be arbitrary, any data that is not an inner product space gives a model of ¬REP-VSLIN, and m is not needed.
   - *No vector space is fixed, even up to isomorphism.* This is false: by Wigner's theorem the Born rule on $P(H)$ fixes H up to a unitary or anti-unitary map (not checked in Lean for two different spaces; the Wigner theorem in `ForPhyslib` is for maps of one space to itself). For $m(H)$ in particular, $m(H) \cong H$ by construction (`twistedIsometry`).
   - *The linear structure is not fixed by the rays and the Born rule.* This is proved (`exists_second_linear_structure`): the carrier of H has a second complex inner product space structure, carried over along m, with the same scalar multiplication, hence the same rays, and the same Born rule, but a different addition. It uses eqs. (3), (4) and (6), and needs $\dim H \geq 2$ (`map_add_of_map_smul`).

   The step of the proof that breaks under the first two readings is "since the map is non-linear, $H \not\equiv m(H)$ in the category of vector spaces", which is false (Log, "Problems found"). The third reading is proposed as the precise statement. The item stays open until the brief adopts it, or another one, and corrects that sentence. (Log, "Differences from the brief".)

2 Related items

From `open-problems.tex`, "From the brief":

4. **Any dimension.** The proof needs a non-linear, Born-rule-preserving, ray-preserving map on the $\mathcal{H}$ of the system, but constructs m only for a two-state system. The same formula $m(\psi) = e^{\imath\theta\, p(z^+|\psi)}\psi$ for a unit vector $|z^+\rangle$ should work whenever $\dim \mathcal{H} \geq 2$, including infinite dimension; this is not formalized ($\mathcal{H} = \mathbb{C}^2$ only). (Brief, Sec. 3, eq. (2); Log, "Setting".)

From `formalization-log.tex`, "Problems found in the brief":

- The closing remark says that m is a diffeomorphism. It is not differentiable at 0: along every real line $m(tv) = t\,m(v)$, so a derivative at 0 would have to equal m itself, and m is not additive. The correct statement is that m is a homeomorphism of $\mathbb{C}^2$ and a diffeomorphism of $\mathbb{C}^2 \setminus \{0\}$. Theorem 1 only needs a bijection, so it is not affected.
- The proof says that "since the map is non-linear, $H \not\equiv m(H)$ in the category of vector spaces". This is false. As a set $m(H) = H$, since m is a bijection. With the vector space structure of H, $m(H)$ is H. With the structure carried over along m, m is by construction a linear isometry $H \to m(H)$ (`twistedIsometry`). What is true is that the identity of the carrier is not linear from one structure to the other.
- Eq. (6) holds for every map that sends each vector into its own ray, since the Born rule only sees rays (`born_smul`). Any $v \mapsto \varphi([v])\,v$ with $\varphi\colon P(H) \to \mathbb{C} \setminus \{0\}$ would do. What matters for the proof is that m commutes with scalars (eq. (3)) and is not additive (eq. (4)).
- QST allows $\dim H \leq 1$. There every map that commutes with scalars is additive (`map_add_of_map_smul`), so there is no second linear structure. Theorem 1 claims only consistency, so the two-state example suffices; "for every system" would need $\dim H \geq 2$.

3 The author's replies (2026-11-02)

Paraphrased.

1. The point is that H and $m(H)$ are still isomorphic, since an isomorphism exists, but m is not an isomorphism. — This is correct when $m(H)$ keeps the operations of H: then m is not linear. It is equivalent to the third reading above: m is linear from H to H exactly when the identity is linear from H to H with the carried-over structure.
2. The formal statement of Theorem 1 is precise but not what matters physically; the counterexample is the content. What it means for a physical system to have a mathematical characterization cannot be expressed formally.
3. Proposal: the briefs get a section of *mathematical prerequisites*, which states what Lean should prove and is fully formalized. The relations between physical conditions are argued informally from these prerequisites and are checked by reading, not by Lean. The author will rewrite the briefs accordingly.

Under this proposal the blocker is no longer a question of which formal reading of ¬REP-VSLIN is right. It becomes: which mathematical statements go into the prerequisites, and which premise links them to Theorem 1.

4 Proposed resolution

Guess. The prerequisites state the construction for every complex inner product space with $\dim H \geq 2$, finite or infinite, and add that it is impossible for $\dim H = 1$. The informal argument uses the premise:

> A mathematical structure of a representation is a physical requirement only if it is fixed by the states and their conditional probabilities (the rays and the Born rule).

Theorem 1 then says, informally: for every system with $\dim H \geq 2$, QST does not fix the linear structure of H, so the linearity in REP-VSLIN is a choice of representation. This is the third reading of Section 1, with the premise made explicit.

Why.

- *The premise is the brief's own.* The abstract says that m "preserves the Born rule, and therefore all the physics". This is operational equivalence: two descriptions that give the same probabilities for every measurement are the same physics. It is also the premise of Wigner's theorem.
- *m changes only unobservable phases.* m multiplies each vector by a phase that depends only on its ray. The overall phase of a vector cannot be measured, so on states m is the identity: a relabelling, not a new theory.
- *Wigner's theorem leaves exactly this gap.* A bijection of $P(H)$ that preserves the Born rule comes from a unitary or anti-unitary map, but this concerns rays. Nothing forces the map on vectors to be linear, and m induces the identity on rays while being non-linear on vectors. The gap exists in every dimension ≥ 2, so the prerequisite should be general.
- *The conclusion is meant for every system.* "Linearity is not part of the physics" is a claim about quantum systems in general, not about qubits. One example suffices for logical consistency, but not for the intended claim.
- *Dimension 1 is not a physical system.* It has one state, its only observables are multiples of the identity, and every probability is 1. Excluding it costs nothing, and it is exactly where the construction fails (`map_add_of_map_smul`).
- *Superposition is not lost.* Interference lies in the probabilities between rays, which m preserves. Superposition in $m(H)$ is the carried-over addition, and every linear operator U on H becomes the operator mUm^{-1}, linear for it. Unlike non-linear dynamics in the style of Weinberg, which allows signalling, this predicts nothing new.

Confidence.

- High: the premise above is what the author means; the prerequisites should hold for $\dim H \geq 2$ in general, with $\dim H = 1$ excluded.
- Medium: the scope is a single system. For composite systems, applying m to each factor while keeping the usual tensor product need not be consistent; the tensor product would have to be carried over along m as well. The brief does not discuss composite systems, so it should either say "a single system" or treat them.

5 Mathematical prerequisites to add to the brief

Each item is a purely mathematical statement for Lean. Let H be a complex inner product space, $z \in H$ a unit vector and $\theta \in \mathbb{R}$. Write $p(z \mid v) = |\langle z|v\rangle|^2/\|v\|^2$ for $v \neq 0$, and define $m_\theta(0) = 0$ and $m_\theta(v) = e^{\imath\theta\, p(z|v)}\, v$ for $v \neq 0$.

P1. *Properties of m_θ.* m_θ is a homeomorphism of H with inverse $m_{-\theta}$, and a diffeomorphism of $H \setminus \{0\}$. It commutes with every complex scalar, sends each vector into its own ray with a factor of modulus 1, and preserves norms and the Born rule. It is not differentiable at 0 when $\dim H \geq 2$ and $\theta \in (0, 2\pi)$.

Lean, for $H = \mathbb{C}^2$ only: `nlMap_smul`, `nlMap_mem_ray`, `norm_nlMap`, `born_nlMap`, `nlHomeomorph`, `contDiffAt_nlMap`, `not_differentiableAt_nlMap_zero`.

P2. *Non-linearity.* If $\dim H \geq 2$ and $\theta \in (0, 2\pi)$, then m_θ is not additive.
Lean, for $H = \mathbb{C}^2$ only: `nlMap_not_add`, `not_isLinearMap_nlMap`.

P3. *A second linear structure.* If $\dim H \geq 2$ and $\theta \in (0, 2\pi)$, the set H carries a second complex inner product space structure, carried over along m_θ, with the same scalar multiplication (hence the same rays) and the same Born rule, but a different addition. m_θ is a linear isometry from H onto it.
Lean, for $H = \mathbb{C}^2$ only: `exists_second_linear_structure`, `twistedIsometry`.

P4. *Dimension one.* If $\dim H = 1$, every map $H \to H$ that commutes with complex scalars is additive, so no second linear structure with the same scalar multiplication exists.
Lean, for every 1-dimensional space: `map_add_of_map_smul`.

Lean work needed: P1–P3 for a general H with $\dim H \geq 2$ (item 4 of Section 2). The formula is the same; the two-state proofs should generalize with z^+ replaced by z and z^- by a unit vector orthogonal to z.

6 Other changes to the brief

1. *State the premise.* Add the premise of the previous section, for instance after Condition REP-VSLIN, as the link between the prerequisites and Theorem 1.

2. *Theorem 1.* Restate it as the informal consequence of P3 and the premise: for a system with $\dim H \geq 2$, the states and the Born rule do not fix the linear structure of H. If "independent" stays, say what ¬REP-VSLIN means in these terms.

3. *The sentence $H \not\equiv m(H)$.* Replace "Since the map is non-linear, $H \not\equiv m(H)$ in the category of vector spaces" by, for example: "With the operations carried over along m, the set H is again an inner product space, isomorphic to H via m; but m is not linear for the operations of H, so the same states and Born rule are represented by two different linear structures."

4. *Dimension.* Replace "Consider a two-state system" by a general H with $\dim H \geq 2$ and a unit vector $|z^+\rangle$, keeping the two-state case and Fig. 1 as the illustration. Note that for $\dim H = 1$ no such map exists (P4).

5. *Diffeomorphism.* In the Remark, "m is ... a diffeomorphism" should read: a homeomorphism of H and a diffeomorphism of $H \setminus \{0\}$ (P1).

6. *The Remark on transport.* It can cite P3: since m is a linear isometry onto the carried-over structure, superpositions, observables and time evolution transfer along m. This settles item 1 of `open-problems.tex` for everything defined from the linear and inner product structure.

7. *Scope.* Say "a single system", or treat composite systems (see Confidence).

8. *Optional.* Note that eq. (6) holds for any map that rescales within rays; what the proof needs is eq. (3) and eq. (4).

When the blocker closes. Once the brief has the prerequisites P1–P4 and the premise, Lean verifies P1–P4, and Theorem 1 is recorded in the log as argued informally from them and checked by reading.